MSDM 5058 Information Science
Assignment 1 (due 28th February, 2026)

Submit your assignment solution on Canvas. You may discuss with others or seek help from your TA, but you should not directly copy from others. Otherwise, it will be considered as plagiarism.

(1) Bayes' theorem

A telegraphic communications system transmits dots and dashes. Assume that due to noise, an average of $2/5$ of the dots and $1/3$ of the dashes are changed. Suppose that the ratio between the transmitted dots and the transmitted dashes is $5:3$.

(a) What is the probability that a received signal will be the same as the transmitted signal if …
   (i) … the received signal is a dot?
   (ii) … the received signal is a dash?

(b) Assuming independence between signals, if the message dot-dot was received, what is the probability distribution of the four possible messages that could have been sent?

Solution:

(a) Let A be the event that a dot is received, and B that a dash is received. One can make two hypotheses: H_1 that the transmitted signal was a dot; and H_2 that the transmitted signal was a dash. By assumption, one has $P(H_1) : P(H_2) = 5 : 3$ and $P(H_1) + P(H_2) = 1$. Therefore $P(H_1) = 5/8$, $P(H_2) = 3/8$. One knows that

$$P(A|H_1) = \frac{3}{5} \qquad P(B|H_1) = \frac{2}{5}$$

$$P(A|H_2) = \frac{1}{3} \qquad P(B|H_2) = \frac{2}{3}$$

The probabilities of A and B are determined from the total probability formula:

$$P(A) = \frac{5}{8}\frac{3}{5} + \frac{3}{8}\frac{1}{3} = \frac{1}{2} \qquad P(B) = \frac{5}{8}\frac{2}{5} + \frac{3}{8}\frac{2}{3} = \frac{1}{2}$$

From Bayes' theorem, the required probabilities are

$$\text{(i)} \quad P(H_1|A) = \frac{P(H_1)P(A|H_1)}{P(A)} = \frac{(5/8) \times (3/5)}{1/2} = \frac{3}{4}$$

$$\text{(ii)} \quad P(H_2|B) = \frac{P(H_2)P(B|H_2)}{P(B)} = \frac{(3/8) \times (2/3)}{1/2} = \frac{1}{2}$$

(b) As dot-dot was received, A happened twice, so the distribution of the four possibilities are

Signal sent	Probability
$H_2 - H_2$ (dash-dash)	$\frac{1}{4}\frac{1}{4} = \frac{1}{16}$
$H_2 - H_1$ (dash-dot)	$\frac{1}{4}\frac{3}{4} = \frac{3}{16}$
$H_1 - H_2$ (dot-dash)	$\frac{1}{4}\frac{3}{4} = \frac{3}{16}$
$H_1 - H_1$ (dot-dot)	$\frac{3}{4}\frac{3}{4} = \frac{9}{16}$

(2) Characteristic function

A random variable X has probability density

$$f(x) = \frac{1}{2}e^{-|x|}$$

(a) Find its characteristic function.

(b) From the characteristic function, find its mean and variance.

Solution:

(a) Let $M(u)$ be its characteristic function.

$$M(u) = \int_{-\infty}^{+\infty} e^{iux} f(x)dx = \frac{1}{2}\int_{-\infty}^{+\infty} e^{iux-|x|}dx$$
$$= \frac{1}{2}\int_{0}^{\infty} e^{(iu-1)x}dx + \frac{1}{2}\int_{-\infty}^{0} e^{(iu+1)x}dx$$
$$= \frac{1}{2}\left(\frac{1}{1-iu} - \frac{1}{1+iu}\right) = \frac{1}{1+u^2}$$

(b) The nth moment of X can be obtained by using

$$\langle x^n \rangle = \int_{-\infty}^{+\infty} x^n f(x)dx = (-i)^n \left.\frac{d^n M(x)}{du^n}\right|_{u=0}$$

Therefore, the mean is

$$\langle x \rangle = -i\left.\frac{dM}{du}\right|_{u=0} = -i\left.\left[\frac{-2u}{(1+u^2)^2}\right]\right|_{u=0} = 0$$

and the variance is

$$\langle x^2 \rangle - \langle x \rangle^2 = (-i)^2 \frac{d}{du}\left.\left[\frac{-2u}{(1+u^2)^2}\right]\right|_{u=0} - 0 = \left.\frac{2-6u^2}{(1+u^2)^3}\right|_{u=0} = 2$$

(3) Autocorrelation

Let $X(t)$ be a stationary random function whose values are normally distributed with zero mean and variance σ^2, then let

$$Z(t) = \frac{1}{2}\left[1 + \frac{X(t)X(t+\tau)}{|X(t)X(t+\tau)|}\right]$$

for some lag τ. Show that the time average of $Z(t)$ is

$$\bar{Z}(\tau) = \frac{1}{\pi}\operatorname{acos}\left[-\rho(\tau)\right]$$

where $\rho(\tau)$ is the normalized correlation function of $X(t)$. *[Hint: Let $x = X(t)$ and $y = X(t+\tau)$ be two normal random variables and consider their joint distribution.]*

Solution:

Let $x = X(t)$ and $y = X(t+\tau)$ be two normal random variables with correlation $\rho = \rho(\tau)$. Their joint distribution is given by

$$f(x,y) = \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{-\frac{x^2+y^2-2\rho xy}{2(1-\rho^2)\sigma^2}}$$

Then the required average can be represented as

$$\bar{Z} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\frac{1}{2}\left[1+\frac{xy}{|xy|}\right]f(x,y)dxdy$$

Since $\frac{1}{2}\left[1+\frac{xy}{|xy|}\right]$ is identically equal to zero if the signs of *x* and *y* are different, and equal to one otherwise, we have

$$\begin{aligned}\bar{Z} &= \int_{-\infty}^{0}\int_{-\infty}^{0} f(x,y)\,dxdy + \int_{0}^{\infty}\int_{0}^{\infty} f(x,y)\,dxdy \\ &= 2\int_{0}^{\infty}\int_{0}^{\infty} f(x,y)\,dxdy\end{aligned}$$

Change variables to r and θ by letting $x = r\cos\theta$ and $y = r\sin\theta$, then let $z^2 = \frac{r^2(1-\rho\sin 2\theta)}{2(1-\rho^2)\sigma^2}$.

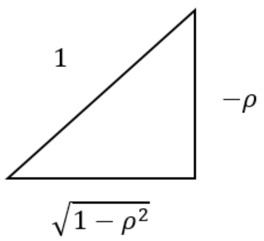
$$\bar{Z} = \frac{2}{2\pi\sigma^2\sqrt{1-\rho^2}}\int_{0}^{\frac{\pi}{2}}\int_{0}^{\infty} e^{-\frac{r^2(1-\rho\sin 2\theta)}{2(1-\rho^2)\sigma^2}}\,rdrd\theta$$

$$= \frac{2}{2\pi\sigma^2\sqrt{1-\rho^2}}\int_{0}^{\frac{\pi}{2}}\left[\frac{(1-\rho^2)\sigma^2}{1-\rho\sin 2\theta}\right]d\theta\int_{0}^{\infty} e^{-z^2}d(z^2)$$

$$= \frac{\sqrt{1-\rho^2}}{\pi}\int_0^{\frac{\pi}{2}} \frac{d\theta}{1-\rho\sin 2\theta} = \frac{1}{\pi}\operatorname{atan}\left(\frac{\tan\theta-\rho}{\sqrt{1-\rho^2}}\right)\Bigg|_0^{\frac{\pi}{2}}$$

$$= \frac{1}{\pi}\left[\frac{\pi}{2} - \operatorname{atan}\left(\frac{-\rho}{\sqrt{1-\rho^2}}\right)\right]$$

Finally, one can convert this result to $\bar{Z}(\tau) = \frac{1}{\pi}\operatorname{acos}\left[-\rho(\tau)\right]$ by considering the following right-angled triangle.



(4) Radar system

A radar system uses radio waves to detect aircraft. After receiving a signal, the system needs to decide whether an aircraft is present or not. Let X be the received signal. We know that

- $X = W$ if no aircraft is present
- $X = 1 + W$ if an aircraft is present

where $W \sim N\left(0, \sigma^2 = \frac{1}{9}\right)$ is some inevitable noise. Thus, we can write $X = \theta + W$, where $\theta = 0$ if there is no aircraft and $\theta = 1$ if there is an aircraft. Now we define the null hypothesis H_0 and the alternative hypothesis H_1 as

- H_0: no aircraft is present
- H_1: an aircraft is present

(a) Write H_0 and H_1 in terms of possible values of θ.

(b) Design a test to decide between H_0 and H_1 by setting its significance level as $\alpha = 0.05$.

(c) Find the probability β of type II error for the test in (b).

(d) If we observe $X = 0.6$, is there enough evidence to reject H_0 at significance level $\alpha = 0.01$?

(e) If we would like the probability of missing a present aircraft to be less than 5%, what is the smallest significance level that we can achieve?

(f) Design another test to decide between H_0 and H_1 by comparing the likelihood ratio of the two hypotheses with some preset threshold η.

Solution:

(a) $H_0 : \theta = 0$ and $H_1 : \theta = 1$. Note that here both hypotheses are simple.

(b) We need to choose a threshold c. If the observed value of X is less than c, we choose H_0 (i.e., $\theta = 0$). If the observed value of X is larger than c, we choose H_1 (i.e., $\theta = 1$). To choose c, we use the required α:

$$\begin{aligned}\alpha &= P(\text{type I error}) = P(\text{reject } H_0|H_0) = P(X > c|H_0)\\ &= P(W > c) = 1 - \Phi(3c)\end{aligned}$$

where Φ denotes the CDF of the standard normal distribution. For $\alpha = 0.05$, $c = 0.548285$. (Note that this test corresponds to the Neyman-Pearson detector of our lecture notes.)

(c) We have

$$\begin{aligned}\beta &= P(\text{type II error}) = P(\text{accept } H_0|H_1) = P(X < c|H_1)\\ &= P(1 + W < c) = P(W < c - 1) = \Phi\big(3(c-1)\big)\end{aligned}$$

For $c = 0.548285$, $\beta = 0.0876857$.

(d) We have got $\alpha = 1 - \Phi(3c)$ in part (b). For $\alpha = 0.01$, $c = 0.775449$, which is larger than $X = 0.6$. Therefore, we cannot reject H_0 at significance level $\alpha = 0.01$.

(e) Here we want $\beta = \Phi\big(3(c-1)\big) \leq 0.05$, which gives $c \leq 0.451715$ and thus $\alpha \geq 1 - \Phi(3c) = 0.0876857$. Therefore, the smallest significance level that we can achieve is $\alpha = 0.0876857$.

(f) Under H_0, $X \sim N\left(0, \frac{1}{9}\right)$; and under H_1, $X \sim N\left(1, \frac{1}{9}\right)$. The resultant likelihood ratio is

$$\Lambda(x) = \frac{f_X(x|H_1)}{f_X(x|H_0)} = \frac{\frac{3}{\sqrt{2\pi}} e^{-\frac{9(x-1)^2}{2}}}{\frac{3}{\sqrt{2\pi}} e^{-\frac{9x^2}{2}}} = e^{\frac{9(2x-1)}{2}}$$

If $\Lambda(X) > \eta$ or $X > \frac{1}{2}\left(1 + \frac{2}{9}\ln\eta\right)$, we accept H_1. If $\Lambda(X) < \eta$ or $X < \frac{1}{2}\left(1 + \frac{2}{9}\ln\eta\right)$, we accept H_0. On the other hand, if you define the likelihood ratio as

$$\Lambda(X) = \frac{f_X(x|H_0)}{f_X(x|H_1)} = \frac{\frac{3}{\sqrt{2\pi}} e^{-\frac{9x^2}{2}}}{\frac{3}{\sqrt{2\pi}} e^{-\frac{9(x-1)^2}{2}}} = e^{\frac{9(1-2x)}{2}}$$

like statisticians and economists, you accept H_1 if $\Lambda(X) < \eta$ or $X > \frac{1}{2}\left(1 - \frac{2}{9}\ln\eta\right)$ but accept H_0 if $\Lambda(X) > \eta$ or $X < \frac{1}{2}\left(1 - \frac{2}{9}\ln\eta\right)$. (Note that this test corresponds to the Bayes detector of our lecture notes.)